WIRELESS COMMUNICATION NETWORKS AND SYSTEMS

LAB 01 REPORT

Fundamentals of Signal Transmission

Python simulation • ***BELHIMER Mohmaed Ayoub***

1. EXPERIMENTAL RESULTS & ANSWERS TO QUESTIONS

Step 1 – Generate a Sine Wave

We generate a sine wave `s(t) = A · sin(2πft)` with `A = 1`, `f = 1000 Hz`, and f_s `= 10000 Hz` over `5 ms`. We also test 500 Hz and 2000 Hz.

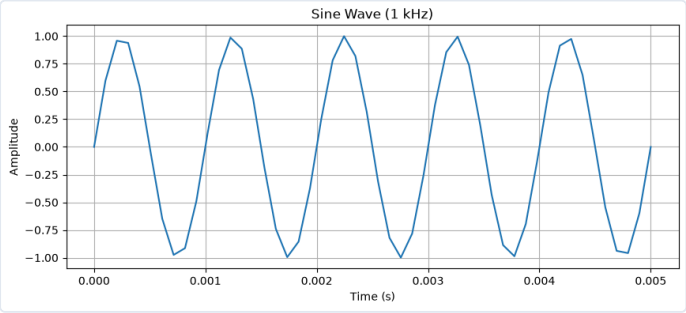


Figure 1.1: 1 kHz sine wave (5 cycles over 5 ms).

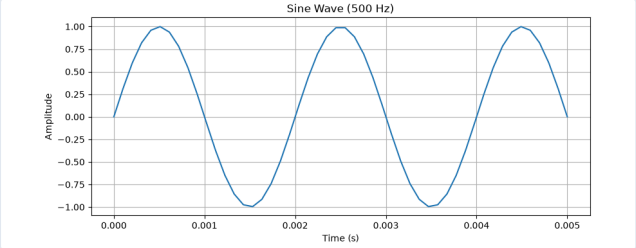


Figure 1.2: Sine wave at 500 Hz (2.5 cycles).

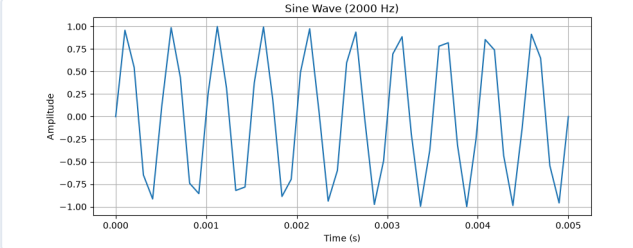


Figure 1.3: Sine wave at 2000 Hz (10 cycles).

Task: Change the frequency to 500 Hz and 2000 Hz. What happens?

Answer: Lower frequency gives fewer cycles in the same time (2.5 cycles for 500 Hz). Higher frequency gives more cycles (10 cycles for 2000 Hz). In both cases, the amplitude stays the same at 1.

Step 2 – Frequency Analysis (FFT)

We compute the FFT to see the signal's frequency components.

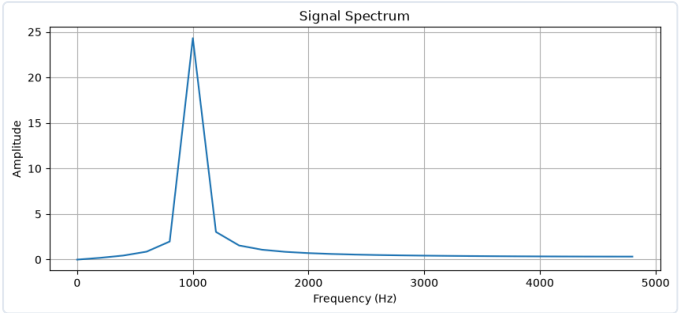


Figure 2: Spectrum of the 1 kHz sine wave.

Question: Identify the dominant frequency in the spectrum.

Answer: The dominant frequency is 1 kHz (1000 Hz), which is where the main peak appears in the spectrum.

Step 3 – Attenuation

We simulate attenuation by scaling the signal: `attenuated = 0.5 * signal`.

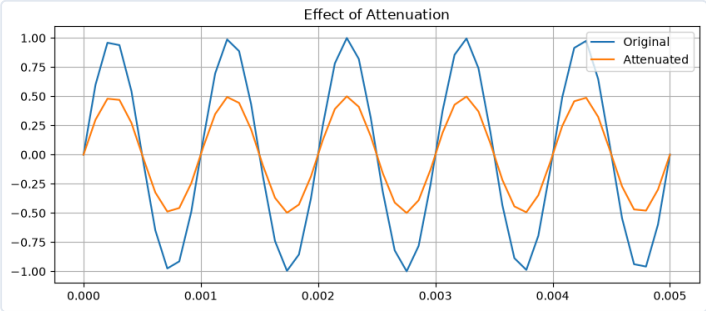


Figure 3: Original vs. attenuated signal.

Question: Why is attenuation problematic for Wi-Fi?

Answer: Attenuation reduces the signal strength as it travels through the air and obstacles like walls. This makes the Wi-Fi signal harder for devices to detect, leading to connection drops, lower data speeds, and reduced coverage range.

Step 4 – Add Noise

We add Gaussian noise with different values of σ (0.1, 0.5, 1.0) and observe the signal in both time and frequency domains.

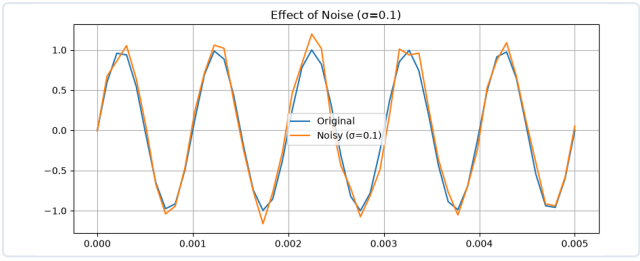


Figure 4.1: Noisy signal (σ = 0.1).

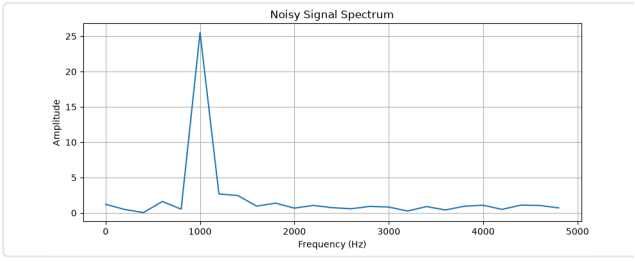


Figure 4.2: FFT spectrum with noise (σ = 0.1).

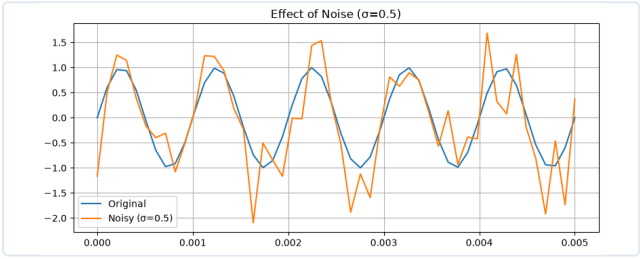


Figure 4.3: Noisy signal (σ = 0.5).

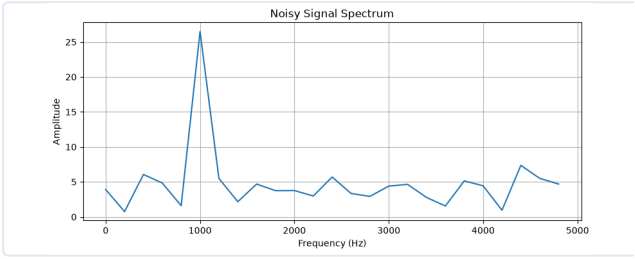


Figure 4.4: FFT spectrum with noise (σ = 0.5).

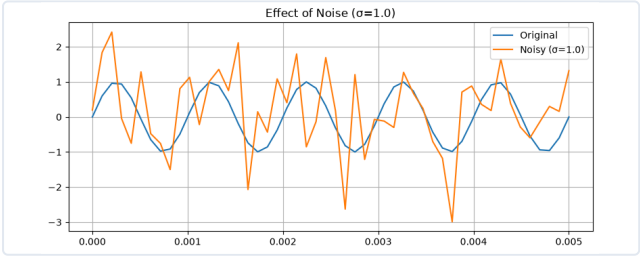


Figure 4.5: Noisy signal (σ = 1.0).

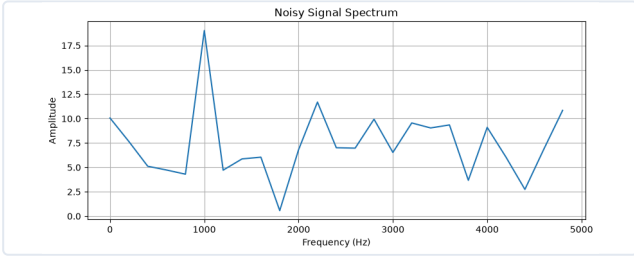


Figure 4.6: FFT spectrum with noise (σ = 1.0).

Task & Question: Vary σ (0.1, 0.5, 1.0) and observe. Why can the FFT still recover the main frequency under noise?

Answer: As σ increases, the time-domain signal becomes much noisier and harder to read. However, the FFT can still show the main frequency because the sine wave concentrates its energy at 1 kHz to form a sharp peak, while the random noise is spread across all frequencies.

Step 5 – Delay

We introduce a time shift by shifting the signal by 20 samples (delay = 20, which is 2 ms).

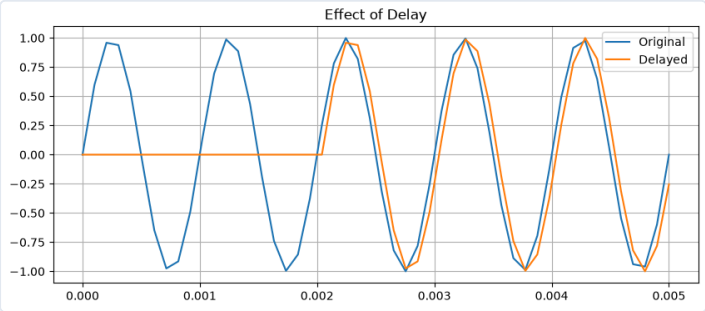


Figure 5: Original vs. delayed signal.

Question: Give a real example where delay is significant.

Answer: Satellite communication is a good example because the signal has to travel very long distances up to the satellite and back, which creates a noticeable delay during phone calls, video calls, or internet browsing.

Step 6 – Combined Channel

We combine all three effects: attenuation (0.6), noise (σ = 0.3), and delay (20 samples).

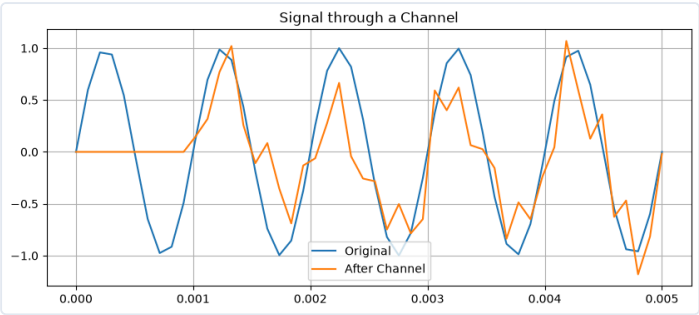


Figure 6: Signal passed through the combined channel.

Question: Compare the original and received signals.

Answer: The received signal is weaker (smaller amplitude), delayed in time (shifted to the right), and distorted by noise compared to the clean original sine wave.

Step 7 – Square Wave Variant

We generate a 1 kHz square wave using `np.sign(np.sin(2*np.pi*f*t))`.

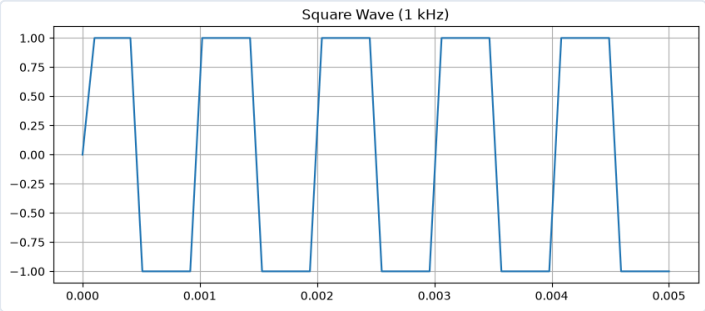


Figure 7: Generated square wave (1 kHz).

Question: Why does a square wave require more bandwidth than a sine wave?

Answer: A pure sine wave is smooth and exists at only one single frequency (1 kHz), so it uses minimal bandwidth. In contrast, a square wave has sharp, instant transitions between -1 and +1. To create these sharp edges, it is made of the base frequency plus many higher odd harmonics (3 kHz, 5 kHz, 7 kHz, ...). Because these higher frequency components spread across a much wider frequency range, transmitting a square wave requires significantly more bandwidth.

2. CONCLUSION

1. In this lab, we simulated the basics of signal transmission and channel effects using Python.
2. We visualized sine waves in time and observed their dominant frequency using the FFT.
3. We tested the three main channel impairments: attenuation weakens amplitude, noise adds random variations, and delay shifts the signal in time.
4. The FFT proved effective under noise because the sine wave peak stays clear while noise spreads across frequencies.
5. Combining all three effects showed how real wireless channels degrade signals by making them weaker, noisy, and delayed.
6. Finally, we saw that square waves require much more bandwidth than sine waves due to their harmonics.